

## A4.2.3 Dimensionality Reduction



### A4.2.3 Describe the importance of dimensionality reduction.

In machine learning, dimensionality refers to the number of features in a dataset. More features might seem to mean more information, but beyond a certain point adding features creates serious problems for model performance, training efficiency, and data analysis. These problems are collectively known as the curse of dimensionality. This page describes each of the seven considerations the syllabus identifies, explains dimensionality reduction as the response, and distinguishes it clearly from feature selection.

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### Key Questions

- What is the curse of dimensionality, and what are the seven considerations the syllabus identifies?
- Why does adding more features eventually harm a model rather than help it?
- Why does the effectiveness of distance-based algorithms decrease in high-dimensional spaces?
- Why does high dimensionality make data visualisation difficult, and why does that matter in practice?
- What is dimensionality reduction, and how does it differ from feature selection?

### IB Command Terms

#### Describe

Give a detailed account of the features or characteristics. For each consideration of the curse of dimensionality: state what the problem is, explain what causes it as dimensions increase, and describe the impact on the model or analysis.

#### What to include

All seven considerations named in the syllabus: overfitting, computational complexity, data sparsity, effectiveness of distance metrics, data visualisation, sample size increases, and memory usage. For dimensionality reduction make sure to define it, explain what it preserves, and distinguish it from feature selection.

## The Curse of Dimensionality

### What is the curse of dimensionality?

In machine learning, a dimension is one feature (one column in the dataset). A dataset describing houses might have 5 dimensions (size, bedrooms, age, postcode, price). A dataset describing medical patients might have 500 dimensions (every measurement, test result, and demographic variable).

The curse of dimensionality is the collection of problems that arise when working with high-dimensional data. These problems are not caused by any flaw in the data or the algorithm they are mathematical consequences of how space behaves as dimensions increase. Understanding them is essential to understanding why dimensionality reduction is needed.

#### THINK FIRST

Imagine you have 100 data points. On a line (1 dimension) they are packed closely together. On a square (2 dimensions) they start to spread out. In a cube (3 dimensions) they are further apart still. What do you think happens as you continue adding dimensions? At what point does the space become so large that 100 points tell you almost nothing about it?

#### 1. Data sparsity

**What it is:** As the number of dimensions increases while the number of data points stays the same, the data points become increasingly spread out. Most of the high-dimensional space contains no data at all. This is called data sparsity.

**Impact on the model:** Algorithms that rely on finding patterns in local neighbourhoods (regions of the feature space where many similar examples cluster) cannot work effectively when data is sparse. If every point is far from every other point, there are no meaningful neighbourhoods to analyse. The algorithm is effectively working in an almost entirely empty space.

**Example:** A dataset of 1,000 patient records has reasonable density in 5 dimensions. In 1,000 dimensions (one per medical test), the same 1,000 patients are isolated points in a space so large that no two patients are meaningfully close to any other.

## 2. Overfitting

**What it is:** With many features, a model has a large number of parameters which means one set of weights per feature, plus interactions between features. This gives the model enormous flexibility. It can describe extremely complex decision boundaries.

**Impact on the model:** When a model has more parameters than necessary, it uses that excess flexibility to memorise coincidental patterns in the training data rather than learning the true underlying relationship. The result is a model that scores very highly on training data but performs poorly on new, unseen examples. This is overfitting. Adding features that are not genuinely predictive gives the model more opportunities to find and memorise noise.

**Example:** A spam filter trained on 10,000 features (every word that ever appeared in a training email) may learn that one specific email address sent spam in the training set and fire on that address alone which is a coincidence, not a pattern. With only 50 carefully chosen features, this kind of memorisation is far less likely.

## 3. Computational complexity

**What it is:** Every feature the model processes requires additional computation. Training involves performing mathematical operations across all features for every example in the dataset, repeated many times until the model converges. The number of operations scales with the number of features.

**Impact on the model:** Training time grows rapidly with the number of dimensions. A model that trains in seconds with 10 features may take hours with 1,000 features and days with 10,000. For iterative algorithms that pass through the data many times, this multiplied cost can make high-dimensional training entirely impractical without specialist hardware.

**Example:** A deep learning model processing medical images of size 512x512 pixels with three colour channels has over 786,000 input dimensions per image. Without dimensionality reduction applied to the feature representation, training on thousands of such images becomes prohibitively expensive.

## 4. Effectiveness of distance metrics

**What it is:** Many ML algorithms including K-Nearest Neighbours (K-NN), K-Means clustering, and Support Vector Machines use distance as their core measure of similarity. Two data points that are close in feature space are considered similar; two points that are far apart are considered different.

**Impact on the model:** In high-dimensional spaces, a mathematical phenomenon occurs: the distance between the nearest neighbour and the furthest neighbour of any given point becomes nearly identical. Every point appears approximately equidistant from every other point. When all distances are essentially the same, 'nearest' loses meaning this means that the algorithm cannot distinguish similar from dissimilar examples, and its predictions become unreliable.

**Example:** *A K-NN classifier with 5 dimensions might correctly identify the 3 most similar patients to a new patient and use their diagnoses to predict. With 500 dimensions, the distances between the new patient and all other patients converge to the same value so the algorithm has no basis for choosing which 3 to use.*

#### WORKED EXAMPLE: DISTANCE METRIC DEGRADATION

Consider a simple dataset where all feature values are drawn from the range 0 to 1. In 1 dimension, a point at 0.2 and a point at 0.8 are clearly far apart (distance 0.6); a point at 0.5 is clearly the nearest neighbour.

As dimensions increase, each dimension adds a small contribution to the total distance. With 100 dimensions, the distances between all pairs of points converge to approximately the same value (around 1.41 for random points in  $[0,1]^{100}$ ). The ratio of the nearest to furthest distance approaches 1. Algorithms using this distance cannot distinguish near from far.

This is not a flaw in the distance formula it is a property of high-dimensional geometry. The only solution is to reduce the number of dimensions so that distance retains discriminative power.

## 5. Data visualisation

**What it is:** Humans can perceive and interpret data in two or three dimensions such as a scatter plot, a bar chart, a 3D graph. Beyond three dimensions, direct visual representation of data is impossible. We cannot draw or imagine a 50-dimensional scatter plot.

**Impact on the model:** Data visualisation is a critical tool in exploratory data analysis: it allows data scientists to spot patterns, anomalies, clusters, and relationships before building a model. In high-dimensional data, this visual inspection is not possible without first reducing the dimensions to a representable number. Problems that would be immediately obvious in a 2D

plot such as a cluster of outliers, a clear separation between classes remain hidden in high-dimensional data.

**Example:** *A dataset of 1,000 customer records with 200 features cannot be directly visualised. After dimensionality reduction to 2 dimensions, the data can be plotted and a data scientist might immediately observe that customers naturally form three distinct groups. This is an insight that was invisible at 200 dimensions.*

## 6. Sample size increases

**What it is:** To achieve adequate data density in a feature space enough data points in each region of the space to learn reliable patterns the number of required examples grows exponentially with the number of dimensions. Doubling the dimensions does not double the required sample size; it may multiply it by an order of magnitude or more.

**Impact on the model:** In practice, datasets cannot be expanded to match the exponential growth in required sample size. Real-world data collection is limited by cost, time, and availability. A medical study cannot simply produce 100 times more patients because a researcher added 100 more features to measure. The consequence is that high-dimensional models always operate with far less data than their feature space requires, making reliable learning impossible.

**Example:** *A fraud detection model trained on 10 features needs perhaps 10,000 transactions to learn well. Expanding to 100 features might require 10 million transactions to maintain the same data density. This is far beyond what most organisations collect in a year.*

## 7. Memory usage

**What it is:** Storing a dataset requires memory proportional to the number of examples multiplied by the number of features. As dimensions increase, the memory required to hold the dataset in RAM increases proportionally. Operations on high-dimensional data such as matrix multiplications, and distance calculations also require working memory that scales with dimensionality.

**Impact on the model:** Large high-dimensional datasets may exceed the available RAM of the computing system. When this happens, the system must read data from slower storage (disk), dramatically increasing computation time. In some cases, the dataset is simply too large to process on the available hardware without first reducing its dimensionality.

**Example:** *A genomics dataset with 500,000 patients and 1 million genetic variants has 500 billion data points. Storing this in RAM at standard precision requires approximately 4*

*terabytes. This is beyond the capacity of most individual machines. Dimensionality reduction to a manageable number of composite features makes the analysis feasible.*

## Dimensionality reduction

Dimensionality reduction is the process of reducing the number of features in a dataset while preserving as much of the important information and underlying structure as possible. It addresses all seven considerations of the curse of dimensionality simultaneously: with fewer dimensions, data is denser, overfitting risk decreases, training is faster, distances are meaningful, visualisation becomes possible, sample size requirements fall, and memory usage decreases.

The key principle is preservation. Dimensionality reduction is not simply discarding information it is finding a lower-dimensional representation of the data that retains the structure and relationships that matter for the task. The goal is a smaller set of features that captures most of the variance and pattern in the original data.

### HOW DIMENSIONALITY REDUCTION WORKS (CONCEPTUALLY)

Dimensionality reduction transforms the original features into a new, smaller set of composite features. Each composite feature is a mathematical combination of several original features. The transformation is designed to maximise the information retained in the composite features while minimising the number of features needed.

#### For example:

A dataset describing people might contain 'height', 'weight', 'arm\_span', and 'leg\_length'. These four features are highly correlated -- taller people tend to be heavier with longer limbs. Dimensionality reduction might combine all four into a single composite feature capturing overall body size, retaining most of the information in one dimension instead of four.

**Note:** the mathematical methods that implement this, including PCA and LDA, are beyond the scope of this course. You are assessed on understanding what dimensionality reduction achieves, not on how any specific algorithm implements it.

## Feature selection vs dimensionality reduction



These two concepts can be easily confused in exam answers. Both reduce the number of features the model works with, but they do so in fundamentally different ways. The table below clarifies the distinction.

| DIMENSION               | FEATURE SELECTION                                                                                     | DIMENSIONALITY REDUCTION                                                                                                            |
|-------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| What it does            | Selects a subset of the original features and discards the rest. The retained features are unchanged. | Transforms the original features into a new, smaller set of composite features. The new features are combinations of the originals. |
| Original features kept? | Yes - the selected features are the same columns that existed in the original dataset.                | No - the output features are newly created. The original features are combined into a reduced representation.                       |
| Interpretability        | High - retained features still have their original meaning (e.g. 'age', 'income').                    | Lower - the new composite features are mathematical combinations and may not have an intuitive real-world meaning.                  |
| When to use             | When some original features are clearly irrelevant or redundant and can simply be dropped.            | When all or most features contain useful information but the total number is too high for efficient learning.                       |
| Example                 | Keep 'square_footage' and 'num_bedrooms'; discard 'listing_ref' and 'prev_owner_name'.                | Combine 'height' and 'weight' into a single composite feature capturing overall body size.                                          |

COMMON EXAM MISTAKE

Describing dimensionality reduction as 'choosing which features to keep' is incorrect as that is feature selection. Dimensionality reduction creates new composite features from combinations of the originals. The original features no longer exist in the reduced dataset; they have been transformed. An exam answer must make this distinction explicit to score full marks on questions that ask you to compare the two approaches.

Summary: the seven considerations

All seven considerations are equally assessable. Use this table to ensure you can name, describe, and explain the impact of each.

| CONSIDERATION                            | WHAT HAPPENS                                                                                                                            | IMPACT ON THE MODEL                                                                                                                                       |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data sparsity</b>                     | Data points become increasingly spread out as dimensions increase, leaving large empty regions of feature space.                        | <i>Algorithms cannot find meaningful local patterns; every point appears equally distant from every other point.</i>                                      |
| <b>Overfitting</b>                       | With many features, the model has excess flexibility and can memorise coincidental patterns in the training data.                       | <i>High training accuracy but poor performance on new data; the model fails to generalise.</i>                                                            |
| <b>Computational complexity</b>          | Every additional feature increases the number of calculations required per training step.                                               | <i>Training time grows rapidly with dimensions; a model that trains in minutes with 10 features may take days with 1,000.</i>                             |
| <b>Effectiveness of distance metrics</b> | In high dimensions, the difference between the nearest and furthest neighbours becomes negligibly small.                                | <i>Distance-based algorithms (K-NN, K-means) lose their ability to distinguish near from far; all points appear equally distant.</i>                      |
| <b>Data visualisation</b>                | Beyond three dimensions, data cannot be directly visualised by humans.                                                                  | <i>Exploratory analysis and model debugging become much harder; patterns that would be obvious in 2D are invisible in 50D.</i>                            |
| <b>Sample size increases</b>             | To maintain adequate data density, the required number of examples grows exponentially with the number of dimensions.                   | <i>In practice, datasets cannot be expanded to match: high-dimensional models always work with far too little data relative to the space they occupy.</i> |
| <b>Memory usage</b>                      | Storing a dataset with more features requires more memory per example; operations on high-dimensional data require more working memory. | <i>Large datasets with many features may exceed available RAM, requiring slower disk-based processing or restricting what models can be used.</i>         |

## Flashcards

Use this Quizlet to learn all the key terms in this section.





## Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

[View this flashcard set](#)

Choose a Study Mode 

## Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

The "Curse of Dimensionality" refers to problems that arise from having:

- A. ☐ Too few data points.

B. ☐ Data that is not normalized.

C. ☒ Too many features (or columns).

D. ☐ Too many duplicate records.

2

As the number of dimensions in a dataset increases while the number of data points stays the same, the data becomes more:

A. ☐ Biased

B. ☒ Sparse

C. ☐ Linear

D. ☐ Dense

3

A model that performs perfectly on its training data but fails on new data is likely suffering from:

A. ☐ Underfitting

B. ☐ Low computational complexity

C. ☒ Overfitting

D. ☐ Data sparsity

4

What is the primary difference between Dimensionality Reduction and Feature Selection?

A. ☐ There is no difference; the terms are interchangeable.

- B. ☐ Dimensionality Reduction always results in a more accurate model.
- C. ☐ Feature Selection is more computationally expensive.
- D. ☒ Feature Selection keeps a subset of original features, while Dimensionality Reduction creates new features.

5

Which of the following is a direct consequence of increased computational complexity from high dimensionality?

- A. ☐ The risk of underfitting increases.
- B. ☐ The model becomes more interpretable.
- C. ☒ The model takes significantly longer to train.
- D. ☐ The data becomes denser.

6

The concept of "distance" between data points becomes less effective in high dimensions because:

- A. ☐ The distance formula changes.
- B. ☒ The distance between any two points tends to become very similar, making "neighbors" rare.
- C. ☐ The data becomes too easy to visualize.
- D. ☐ All data points collapse into a single point.

7

The main goal of dimensionality reduction is to:

- A. ☒ Reduce the number of features while preserving important information.
- B. ☐ Eliminate all outliers from the dataset.
- C. ☐ Make the data perfectly linear.
- D. ☐ Increase the number of features to make the model more complex.

8

Which of these is NOT a problem associated with the Curse of Dimensionality?

- A. ☐ Overfitting
- B. ☐ Increased memory usage
- C. ☐ Data sparsity
- D. ☒ Underfitting

9

To avoid the curse of dimensionality, if you significantly increase the number of features in your dataset, you should also ideally:

- A. ☐ Remove the bias term from your perceptrons.
- B. ☒ Exponentially increase your sample size (number of data points).
- C. ☐ Decrease the number of hidden layers in your neural network.
- D. ☐ Use a smaller stride in your CNN.

10

A data scientist transforms a dataset with 200 features into a new dataset with 10 features that are combinations of the original 200. This process is an example of:

A. ☐ Dimensionality reduction

B. ☐ Standardization

C. ☐ Data cleaning

D. ☐ Feature selection

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